Avi Gobrin

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Education

University of Toronto Sep 2023 - Apr 2027
Honours BSc, Mathematics & Its Applications Specialist, Statistics Major, Computer Science Minor *Toronto, ON*

Relevant Coursework: Algorithm Design & Analysis, Data Structures, Software Design, Theory of Computation, Statistical Machine Learning, Artificial Intelligence, Nonlinear Optimization, Mathematical Modelling, Real Analysis, Combinatorics, Graph Theory, Statistical Computation

Experience

Nextdoor May 2026 - Aug 2026
Machine Learning Intern *San Francisco, CA*

- Built Best Places for Families end to end as the sole engineer, an SEO ranking product scoring 350,000 US neighbourhoods: Airflow ingestion, a Python scoring service, and a Postgres/GraphQL API to a React page, on Kubernetes (AWS).
- Designed the scoring methodology: screened 33 candidate metrics drawn from 10+ Census/ACS and internal tables down to 8 features, benchmarked OLS, ridge, NNLS, and monotone GBM, then hand-tuned the final weights when no fitted model matched human judgment.
- Built the evaluation harness for a problem with no ground truth: 100+ preregistered adversarial neighbourhood pairs resolved by pairwise Bradley-Terry judgments, cross-checked by LLM and manual review.
- Traced an early correlation between the score and wealth to a geographic join missing a county guard that had leaked out-of-state Census data into 94.6% of neighbourhoods; the fix removed 95% of bad links.
- Made it self-sustaining with scheduled jobs, tests, and Datadog monitoring; built Databricks tooling that lets the PM and design team check any city's ranking and retune weights; presented it company-wide.

Shopify May 2025 - Aug 2025
Software Engineering Intern *Toronto, ON*

- Built a full-stack partner-attach system (Ruby on Rails, React, Calendly API) that screens each new merchant, matches those needing a specific language to a partner who speaks it, and assigns the rest to the least-loaded advisor, routing 100+ vetted merchants per day into guided onboarding that replaced an unassisted migration path.
- Delivered Shopify's German VAT compliance program for the entire German merchant base under a hard legal deadline: connected legacy databases to a new compliance flow, built an intake form for the data Shopify did not hold, and integrated the German tax authority's API to generate and email compliance PDFs.

Projects

Tiny GPT From Scratch *Personal Project* [Repo] 2026

- Implemented a character-level GPT in pure NumPy with no autograd: multi-head causal attention, pre-LN Transformer blocks, Adam, and top-k sampling, with hand-written backward passes for every layer.
- Verified gradients for every model parameter against finite differences in a 56-test pytest suite, run in CI on every push.

Autonomous Planning & School Knowledge Agents *Personal Project* 2026

- Cut daily planning from about 30 minutes to under 2 with an autonomous planning agent (Claude Code, Obsidian) that runs at the start and end of each day, reviewing what was completed, writing structured notes on the day, and regenerating a time-blocked calendar for the next day from a parsed task file and a traffic-light priority scheme.
- Built a companion school-workflow agent that auto-formats lecture and problem-set notes into a cross-linked knowledge base of theorem references, surfacing relevant prior material for homework and exam preparation.

Research

University of Toronto, Department of Statistical Sciences Aug 2026 - Present
Undergraduate Researcher, Advisor: Prof. Ricardo Baptista (Vector Institute) *Toronto, ON*

- Investigating a unifying rank-reduction framework linking in-context learning and fine-tuning as low-rank weight updates, connecting recent transformer interpretability work to classical multivariate statistics.

Technical Skills

Languages: Python, Ruby, Java, JavaScript, SQL, Bash, HTML/CSS, R
Systems & Tools: Ruby on Rails, React, Node.js, Flask, GraphQL, REST APIs, Postgres, AWS, Docker, Kubernetes, CI/CD, Airflow, Databricks, Datadog, Git, Claude Code, LLM agents
Interests: StarCraft II Grandmaster (top 100 in North America), competitive Magic: The Gathering, karate black belt